Graph Smile-Extrapolation

Filling dark nodes from lit neighbours by an OT-regularized Bayesian graph

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Abstract

This is the Vol-Fitter's headline differentiator: extrapolating sparse smile observations to a *full universe* of smiles, across expiries and assets, by propagating signal through a graph whose nodes are smiles (underlying, T). The mathematics is a Gaussian inverse problem with an optimal-transport-regularized prior on the *increments*, not the levels — so the default answer is "nothing changed from the prior", and observed nodes force only the coherent, graph-plausible changes that explain them. We build the directed graph (a row-normalized kernel, its stationary distribution, reversible and directed Laplacians), assemble the increment precision $Q_\Delta = D_\kappa + \eta L_{\mathrm{dir}} + \lambda(A_\rho + \nu I)^{-1}$ from three beliefs (local smallness, directed prediction, low-energy unbalanced-OT flux), and condition on the lit nodes in covariance form exploiting $n_{\mathrm{obs}} \ll N$. We then describe the production spine — transported prior, lit-calibration innovation, graph posterior increment, dark-node reconstruction, quote comparison — with its provenance hierarchy, data-derived precision, and per-edge directed amplitude β. A real run of the production solver shows a lit innovation flowing to dark nodes and decaying with a widening credible band; and a temporal leave-one-out backtest measures the graph's *skill* over the transported-prior baseline at +26 to +37 ATM vol bps, well calibrated (standardized-residual mean ≈ 0).

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1 The problem and the idea

Most of a desk's universe is illiquid. A handful of nodes — index expiries, a few mega-caps — are richly quoted (*lit*); the rest — single-name expiries, far tenors — are barely quoted or not at all (*dark*). The dark nodes still need surfaces. The Vol-Fitter builds them by treating the universe as a *graph*: nodes are smiles (underlying, T), edges encode that one node informs another (calendar edges within a name; cross-asset edges from an index or sector ETF to its single names). A prior surface ties the whole graph together; today's lit calibrations perturb it; the perturbation propagates to the dark nodes.

Heuristic

The central modelling choice is to regularize the *increment* $z = x^1 - x^0$ — how each node moved from its prior — rather than the level x^1. Why? Because the prior already encodes the shape of every smile; what the graph must infer is only the *change*, and changes are far more coherent across the graph than levels (an index move drags its names; a calendar shift is smooth in T). Regularizing increments means the default answer is "no change", and data is required only to justify departures from the prior — exactly the right inductive bias for a sparsely-observed universe.

The carrier is compact: each node propagates its three ATM handles $(\sigma_0, s_0, \kappa_0)$ (Note 01) as independent Gaussian coordinates, and the reconstruction retargets those handles to an arbitrage-free smile in the chosen model. The production spine is fig. 1.

Figure 1: The production spine: a transported prior plus the lit-node calibration innovation $d = \text{calibrated} - \text{transported}$ propagate to a posterior increment, which reconstructs the dark smiles, scored against any quotes.

2 The directed graph

Edge weights w_{ij} read "j informs i". Row-normalizing gives a transition kernel K (row-stochastic), with stationary distribution π ($\pi^\top K = \pi^\top$) and reversibilized conductances $c_{ij} = \pi_i K_{ij}$. From these come two Laplacians used by the prior:

- the **reversible** Laplacian $L_{\text{rev}} = BCB^\top$ (the symmetric part, a graph smoothness energy);
- the **directed residual** $L_{\text{dir}} = (I - K)^\top \Pi (I - K)$, which penalizes a node deviating from the π-weighted prediction of its in-neighbours — the directed prediction rule.

Calendar edges (one name across T) are high-conductance and vol-normalized ($\beta_{\text{vn}} = \sqrt{T_{\text{to}}/T_{\text{from}}}$); cross-asset edges (index→name, ETF→name, same-sector name→name) carry smaller trust.

3 The increment prior

The increment z is given the Gaussian prior with precision

$$\boxed{Q_\Delta = D_\kappa + \eta\, L_{\mathrm{dir}} + \lambda\,(A_\rho + \nu I)^{-1},} \tag{1}$$

encoding three complementary beliefs:

1. $D_\kappa = \mathrm{diag}(\kappa)$: increments are *locally small* (a node does not move far from its prior on its own).
2. ηL_{dir}: increments *respect the directed prediction* — a node's move should be explained by its in-neighbours' moves.
3. $\lambda(A_\rho + \nu I)^{-1}$: increments should be *producible by a low-energy graph flux* plus ν-priced sources/sinks — the unbalanced optimal-transport tangent norm, with A_ρ the mobility-weighted Laplacian and ν the source allowance. This term says "a coherent move is one that could have been transported along the edges cheaply".

Q_Δ is positive-definite ($D_\kappa > 0$, the others PSD). At this scale (universes of $O(10^3)$ nodes) the matrix is formed and inverted densely; the matrix-free sparse path is the deferred performance follow-up (appendix B).

Heuristic

Read the three terms as three answers to "what is a plausible change?" κ: a small one. η: one my neighbours predict. λ: one that could flow cheaply across the graph. Turning λ on (off by default) adds the transport view, which matters when changes should move *mass* along edges rather than just smooth out; ν controls how much un-transported source is allowed before that becomes expensive.

4 Conditioning on the lit nodes

Given a finite baseline $\bar{x}^0$ with precision p^0, the predictive moments are $\mu^- = \bar{x}^0 + m_\Delta$ and $K^- = P_0^{-1} + K_\Delta$ with $K_\Delta = Q_\Delta^{-1}$. Conditioning on lit observations y (precision r) at nodes S is done in *covariance form*, exploiting $n_{\mathrm{obs}} \ll N$: only the observed columns of K^- and a small $n \times n$ innovation system are needed,

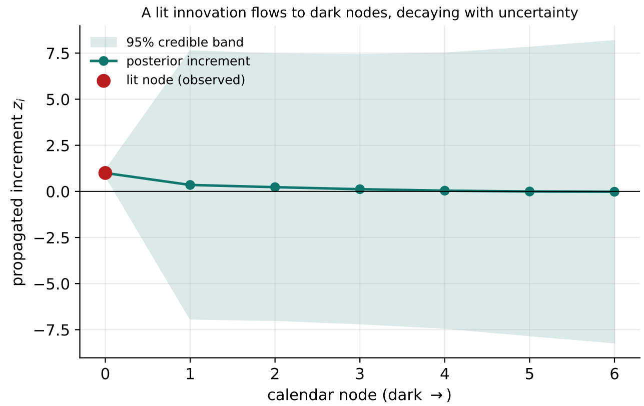
$$S_y\,\alpha = y - \mu^-_S, \qquad \mu^+ = \mu^- + K^-_{\cdot S}\,\alpha, \qquad S_y = K^-_{SS} + \mathrm{diag}(1/r). \tag{2}$$

The marginal posterior variance is $K^+_{ii} = K^-_{ii} - \left(K^-_{\cdot S} S_y^{-1} K^{-\,\mathsf{T}}_{\cdot S}\right)_{ii}$, and the reported per-node precision is $\pi^+_i = 1/K^+_{ii}$.

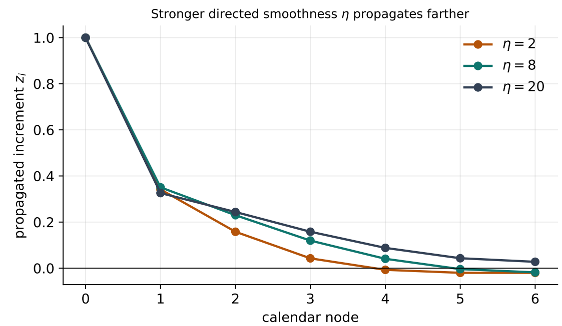
Caution

The reported precision is $1/K^+_{ii}$ — the reciprocal of the marginal *variance* — *not* the i-th diagonal of the posterior precision matrix Q^+. Those differ whenever nodes are correlated (always), and confusing them is the classic Gaussian-graph error: Q^+_{ii} overstates how much a node is pinned, because it ignores that its neighbours are uncertain too. The credible bands in fig. 2 use the marginal variance.

Figure 2 is a real run of the production solver on a calendar chain with one lit node: the unit innovation propagates to the dark nodes, decaying along the graph, while the posterior precision falls from 200.06 at the lit node to 0.06 at the far dark node — the credible band widening accordingly. Section 4 shows the directed-smoothness weight η controlling how far the signal reaches.



(a) propagation + credible band



(b) reach vs η

Figure 2: A lit innovation propagating through the production graph solver to dark nodes, decaying with a widening credible band; stronger directed smoothness η propagates farther.

5 The production spine

5.1 Transported prior and provenance

Each node's baseline x^0 is the prior's three handles, *transported to the current spot* (Note 12). The prior is resolved by a strict hierarchy, each level carrying its provenance and a validity flag: *active transported* (a saved prior) → *nearest-expiry transported* (reduced precision) → *today bootstrap* (weak, not valid for validation) → *flat ATM* (diagnostic only). The provenance sets the baseline precision tier.

5.2 Lit innovation, not manual shift

For each lit node the innovation is $d =$ calibrated handles − transported prior handles — the genuine market-versus-prior move, computed by actually fitting the lit node (in pure-market mode, so the innovation is not itself prior-anchored). Dark nodes are never observations.

5.3 Reconstruction and comparison

The posterior handles at a dark node are retargeted (Note 01's `ortho.retarget`) to an exact arbitrage-free smile in the chosen model (LQD exactly; SVI/SIV fitted to it; local-vol as a projection target), carrying the credible band onto the curve. Where a dark node *does* have a few quotes, they are compared against the reconstruction — weighted RMS, inside-spread hit rate, and a standardized residual ζ using the posterior uncertainty — giving an honest, never-trained-on measure of the extrapolation.

5.4 Per-edge amplitude

Trust and amplitude are separate. The edge *weight* is how much node j informs node i; the per-edge *beta* β_{ij} scales the directed *increment* amplitude, $z_i \approx \sum_j K_{ij}\beta_{ij}z_j$, entering the directed residual as $L^{\beta}_{\text{dir}} = (I - K \circ B)^{\mathsf{T}}\Pi(I - K \circ B)$ (PSD by construction; $\beta \equiv 1$ is byte-identical to the plain rule). Beta is per-handle and per-direction — an index ATM move may translate to a name move at 0.7 while the skew translates at a different rate.

6 Does it work? The leave-one-out backtest

The graph's value is its *skill* over the transported-prior baseline — does propagating the lit innovation beat just keeping the prior? A temporal leave-one-out harness measures it: per consecutive captured day-pair, freeze day-($T{-}1$) as the prior, transport it, form the lit innovations, propagate, and compare the graph posterior for *held-out* nodes against their actual day-T calibration — versus the pure transported-prior baseline.

- **Fill a sparse node from its lit neighbours** (withhold each clean node): the graph *decisively* beats transport — ATM skill +37 vol bps (sticky-moneyness transport) to +26 (sticky-strike), wing +3 to +7 bps, with standardized-residual mean ≈ 0 (unbiased) and std 0.72–0.90 (well calibrated, slightly conservative). The two stickiness regimes *bracket* the true skill exactly as posed. The driver is the calendar coupling.
- **Cross-asset to fully-dark names**: in the 8-asset pilot this adds ~ 0 — but for two *measured* reasons, not a verdict against the method: the transported prior is already an excellent same-name predictor at very high baseline precision, and the pilot is starved of the same-sector clusters and sector ETFs that light the cross-asset edges. The follow-ups (a 25-asset capture; a lower baseline precision for dark nodes) are designed to give it a fair test.

So the headline use case — reconstructing a missing or sparse node from its lit neighbours — works, and the experiment honestly reports where it cannot yet exercise the cross-asset path.

7 What is original here

The Gaussian-graph machinery is classical; the originality is the synthesis for volatility. *Regularizing increments, not levels* (1) is the right bias for a sparse universe and makes "no change" the default. The *three-belief OT-regularized prior* unifies local smoothness, directed prediction, and transportable flux in one precision. The *production spine* — transported prior, market-true innovation, provenance-tiered precision, handle-carrier reconstruction — turns the theory into smiles with credible bands. The *per-edge per-handle beta* separates trust from amplitude. And the *leave-one-out skill measurement* — bracketed by the two stickiness regimes, with calibrated standardized residuals — is an honest verdict rather than an in-sample fit. To our knowledge, propagating arbitrage-free smile handles through an OT-regularized Bayesian graph to reconstruct an unquoted universe is novel.

A Hyperparameter atlas

Table 1: Graph-extrapolation hyperparameters.

Knob	Default	Role
Surfaced		
`graphKappaScale` κ	1.0	Local temporal precision (increment smallness).
`graphEtaScale` η	1.0	Directed-smoothness weight.
`graphLambdaScale` λ	0.0	OT tangent-norm weight (off by default).
`graphNu` ν	0.1	Unbalanced-OT source allowance.
`crossBeta` / `edgeBetas`	1.0	Per-edge directed increment amplitude.
edge weights	lattice / user	Calendar / cross-asset trusts (or an explicit edge list).
Hidden (precision tiers)		
`HANDLE_CONFIDENCE`	$[1, 1, 0.01]$	Per-handle (ATM/skew/curv) relative confidence.
`SOURCE_BASE`	tiered	Provenance precision (active $\gg$ nearest $\gg$ bootstrap $\gg$ flat).
`OBS/BASE_PRECISION_FLOOR/CAP`	per-handle	Floors/caps so no fit dominates / no prior vanishes.

B Performance notes

1. **Covariance-form conditioning.** Only the n_{obs} observed columns of K^- and an $n_{\mathrm{obs}} \times n_{\mathrm{obs}}$ innovation solve are formed — cheap when few nodes are lit.

2. **Dense at $O(10^3)$.** Q_Δ and $K_\Delta = Q_\Delta^{-1}$ are formed densely; fine up to $\sim 10^3$ selected nodes. The deferred Phase-10 sparse path replaces the two dense $N \times N$ inverses and the $O(7\, n_{\mathrm{obs}} N^3)$ autotune with sparse solves $(A_\rho + \nu I)u = v$ and a Hutchinson diagonal estimator, for universes $\gg 10^3$ nodes.

3. $\beta \equiv 1$ **golden.** The per-edge beta machinery is byte-identical to the plain directed rule at $\beta = 1$, so it is strictly additive.

4. **Empirical-Bayes autotune.** Hyperparameters $(\kappa, \eta, \lambda, \nu)$ can be set by marginal-likelihood maximization with held-out calibration; this is the $O(7\, n_{\mathrm{obs}} N^3)$ cost the sparse path targets.

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